Shipment Delay Escalation SOP

SYN-SOP-01 | Version 1.0 | Status: Synthetic MVP guidance

Synthetic operational knowledge created for the Option C Logistics project demonstration. This document is not a real company policy and must not be used for live operational decisions.

Purpose	Define when a predicted shipment delay should be monitored, investigated, or escalated.
Applies to	Supply-chain disruption agent demonstration, operations review, and RAG retrieval testing.
Decision authority	Human operations reviewer. The AI system provides evidence and recommendations only.
Evidence rule	Use only observed tool evidence, model output, and retrieved guidance. Do not invent operational facts.

1. Trigger and scope

Apply this procedure after the ML service returns a valid delay probability and the shipment record has been retrieved using the public SHP- reference.

Risk level	Probability	Required response
LOW	< 0.40	Monitor; do not trigger full investigation.
MEDIUM	0.40 to < 0.70	Investigate contributing factors and collect evidence.
HIGH	>= 0.70	Escalate promptly and prepare proportionate mitigation options.

2. Required evidence

- Model delay probability, configured threshold, risk level, and whether investigation is required.
- Leakage-safe shipment facts: product, category, destination, shipping mode, scheduled shipping days, quantity, and order date.
- Historical product, market, and shipping-mode delay rates.
- Destination weather evidence with source, date window, and limitations.
- Supplier availability and inventory coverage when reliable sources exist.
- Retrieved operational guidance with source filenames.

3. Escalation workflow

1. Validate the shipment reference and confirm the prediction came from the approved preprocessing and XGBoost artifacts.
2. For LOW risk, record the result and continue routine monitoring.

3. For MEDIUM risk, collect shipment, historical-delay, weather, supplier, and inventory evidence that is available.
4. For HIGH risk, notify the designated operations reviewer and prepare proportionate mitigation options.
5. Document unavailable evidence and avoid substituting assumptions.
6. Require human approval before rerouting, changing suppliers, or altering customer commitments.

4. Evidence-to-action guardrails

- A probability slightly above the threshold triggers investigation, not automatic rerouting.
- Low-severity weather alone does not justify immediate route changes.
- Missing supplier or inventory information must be stated as a limitation.
- Post-outcome fields such as actual shipping days, delivery status, and late-delivery target must not be exposed to the agent.

5. Required record

Shipment reference	Public SHP- identifier.
Prediction	Probability, threshold, risk level, and model version.
Tool trace	Ordered list of successful and failed tool calls.
Knowledge sources	Retrieved document filenames and relevant excerpts.
Decision	Human-approved action, owner, and review time.